



Limitations and future work Several limitations exist for our approach. We generated high-quality

ultrasound images using a GAN-based approach; however, the current model does not incorporate physics-based consistency between consecutive frames. Additionally, the patient models—comprising 3D label maps and CT scans—remain static and do not account for soft tissue deformation. The results were obtained from a limited number of patient samples and have not yet been scaled to represent a broader population. Future research directions include improving ultrasound simulation quality and noise diversity, modeling soft tissue deformation, scaling to larger patient populations, improving generalization over different patients, and validation with real robotic ultrasound systems in clinical settings.

these are the future works of sonoGym which works have tried solving their limitations.

Direct answer: several recent papers and datasets have already started addressing each of SonoGym's listed limitations — temporal/physics consistency of generated US frames, soft-tissue deformation, limited patient diversity, noise / image realism, and real-robot validation — using approaches you can reuse or build on. Below I list the most relevant works grouped by the specific limitation they target, with a short note on what they do and how they map to SonoGym's future directions. Each sentence that reports a paper or dataset is cited.

Temporal / physics consistency between consecutive frames

- Diffusion / video models for ultrasound: temporal diffusion or transformer-based video models have been used to generate temporally consistent ultrasound sequences (improving frame-to-frame coherence compared to single-image GANs).^{[1] [2]}
- Temporal representation and contrastive learning: methods that explicitly enforce temporal consistency (e.g., temporal masking / contrastive losses over video frames) improve coherence for US video tasks and can be adapted to conditional CT→US video generation.^{[3] [1]}

Notes: replacing per-frame pix2pix with a conditional video diffusion model or a recurrent/temporal GAN (optical-flow / consistency losses) can directly address SonoGym's missing physics/temporal consistency.^{[2] [1]}

Soft-tissue deformation / biomechanics (dynamic patient models)

- Physics / biomechanics simulation for soft tissue: several works integrate nonlinear biomechanical models (finite-element or data-driven approximations) to simulate soft-tissue deformation for interactive surgical simulation.^{[4] [5]}
- SOFA and related frameworks (e.g., LapGym / SOFA-based simulators) provide deformation-capable simulation stacks used in surgical RL environments and can be combined with CT-derived anatomy to model tissue motion under probe/drill contact.^[5]

Notes: adding a deformable torso/soft-tissue layer (FEA or reduced-order surrogate) and re-computing 2D CT/label slices under deformation would address SonoGym's static patient limitation.^[5]

Improving image realism and noise diversity (scaling sim-to-real)

- Physics-inspired diffusion models and hybrid models: diffusion-based ultrasound generation or physics-aware generative models (combining ray-tracing / acoustic priors with learned noise) show improved realism and generalization to real US than single-image GANs.^{[6] [7] [4]}
- Multi-network ensemble / domain randomization: training multiple generative networks (and sampling among them) or applying strong CT→US augmentations improves generalization to unseen imaging noise and has been proposed and used in SonoGym and related works.^[8]
- Large curated datasets: UltraBones100k and similar large ex-vivo datasets expand anatomical and imaging variability and help train more robust simulation or translation networks.^[7]

Notes: replacing pix2pix with conditional diffusion (or physics + diffusion hybrids), augmenting with multiple trained networks, and pretraining on large datasets (UltraBones100k / similar) addresses noise diversity and sim-to-real gaps.^{[6] [2] [7]}

Scaling to larger patient populations and inter-patient generalization

- Large datasets and automated labeling: UltraBones100k (and related releases) provide scaling to tens of thousands of US images with automated bone annotations, which has been shown to improve generalization across anatomy.^[7]
- Data augmentation and domain adaptation: works using domain randomization, multi-seed generator ensembles, and domain adaptation techniques (contrastive pretraining, feature alignment) have been applied to reduce drop in performance on unseen patients.^{[1] [8]}

Notes: the path to scale is combining larger datasets (UltraBones100k, paired CT-US where available), data augmentation, and domain adaptation to reduce inter-patient gaps seen in SonoGym.^{[8] [7]}

Validation with real robotic ultrasound systems (sim-to-real / on-robot tests)

- Sim-to-real for ultrasound/robotics: there are recent preclinical and bench-top studies that moved policies from sim to robotic ultrasound setups, including ultrasound-guided drilling and reconstruction pipelines, and works that evaluate policies on real hardware for navigation and reconstruction.^[9]
- Safe RL and safety filters: methods such as SafeRPlan / state-wise constrained RL have been applied to intraoperative planning to add safety guarantees prior to robot deployment.^{[1] [8]}

Notes: combining safe RL filters and on-robot bench evaluations (as in recent robotic US drilling/reconstruction studies) is the established route to real-robot validation SonoGym suggests.^[1]

Concrete papers / resources you can read or reuse (representative)

- UltraBones100k — large ex-vivo bone US dataset that helps scale training and improve generalization.^[7]
- Temporal diffusion / video US generation works (cardiac US video generation; conditional video diffusion + temporal transformer) for frame consistency.^{[2] [1]}
- Physics-informed / diffusion hybrid ultrasound simulation (diffusion as sound propagation, physics-inspired models).^{[4] [7]}
- Soft-tissue deformation simulation and SOFA-based surgical simulators (LapGym / SOFA examples) to add deformation to patient models.^[5]
- Recent SonoGym-related resources and codebase (project site and repo) show existing mitigations (multi-network LB training, domain randomization) and are a good starting point.^[8]

Example concrete integration plan (1–2 lines each)

- Temporal consistency: train a conditional video diffusion model (CT slices + previous US frames as conditioning) to output temporally consistent US sequences; use optical-flow losses or contrastive temporal losses during training.^{[2] [1]}
- Deformation: couple a lightweight biomechanical model (reduced FEA or learned deformation field) of skin/soft tissue to the rigid CT anatomy and update slices per timestep; reuse SOFA or reduced surrogate models for speed.^[5]
- Data & generalization: pretrain generators and segmentation models on UltraBones100k and mix multiple generator seeds / domain augmentations during RL training to improve inter-patient robustness.^{[8] [7]}
- Real-robot validation: first validate on ex-vivo specimens and bench-top robotic rigs (as in recent pedicle-drilling studies), using safety filters (SafeRPlan) before moving to in-vivo trials.^[1]

If you want, I can:

- extract the most relevant excerpts from any of the cited papers (PDFs) and summarize methods and losses to implement temporal consistency; or ^{[2] [1]}

- propose a concrete architecture (inputs/losses/training recipe) to replace pix2pix with a CT-conditioned video diffusion model tailored to SonoGym's pipeline; or ^[7] ^[2]
- draft an implementation plan to add deformable patient models (SOFA + surrogate FEA) into SonoGym and estimate computational costs. ^[5]

Which of those follow-ups would you like me to do next?

✱

1. <https://arxiv.org/html/2509.01433v1>
2. [https://eprints.whiterose.ac.uk/id/eprint/228681/1/Cardiac Ultrasound Video Generation Using a Diffusion Model with Temporal Transformer_Camera_ready_Wenbin Wang and Ping Lu.pdf](https://eprints.whiterose.ac.uk/id/eprint/228681/1/Cardiac%20Ultrasound%20Video%20Generation%20Using%20a%20Diffusion%20Model%20with%20Temporal%20Transformer_Camera_ready_Wenbin%20Wang%20and%20Ping%20Lu.pdf)
3. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12604299/>
4. <https://www.sciencedirect.com/science/article/pii/S1566253525009583>
5. <https://www.sciencedirect.com/science/article/abs/pii/S1746809425014454>
6. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12031589/>
7. <https://pubmed.ncbi.nlm.nih.gov/40472504/>
8. <https://openreview.net/forum?id=SRStCEJ0Qu>
9. <https://neurips.cc/virtual/2025/poster/121645>
10. SonoGym_compressed-1.pdf
11. <https://collab.dvb.bayern/spaces/TUMdlma/pages/1078573730/Deep+Learning+in+Ultrasound+Elastography>